ZhuLong: An Execution-Grounded LLM Agent for EDA Scripting

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Abstract

Design engineers lose the most time orchestrating flows and patching the gaps where closed EDA tools refuse to interoperate. We present **ZhuLong**, an execution-grounded LLM agent for PyAether, SKILL, and Tcl that closes a generate–execute–refine loop over unified MCP tools (API retrieval, documentation, sandbox execution) and is already deployed in production EDA flows, with a *grounding-gap framework* positing that EDA-scripting correctness is bottlenecked by *observation fidelity* at the L4 (situated) layer, which model scale alone cannot reach—the binding constraint in our setting. On **EDA-Scripting-Bench** (449 tasks, three languages, four commercial runtimes), ZhuLong reaches **76.9%** pooled Pass@1 versus **19.7%** (pure LLM): the system-level observation axis spans 57.2 pp with the model fixed, exceeding the 23.7 pp backbone range with the harness fixed—two axes answering different questions, making observation the binding constraint in the current model-capability regime.

CCS Concepts

• **Hardware** → **Software tools for EDA**; • **Computing methodologies** → *Artificial intelligence*; • **Software and its engineering** → *Software creation and management*.

Keywords

LLM Agents for EDA, EDA Scripting, Code Generation, Execution Grounding, Benchmarking

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1 Introduction

Design engineers' daily work is less about fresh scripts than about orchestrating design flows and patching the gaps between EDA tools that refuse to interoperate—a flow that cannot run end-to-end is the single largest source of lost design time. Commercial EDA tools are closed: where they leave a gap, the only recourse is hand-written glue (a Makefile target, a Tcl driver, a SKILL wrapper), precisely the small intent-to-code synthesis at which LLMs excel. This reframes EDA scripting as the connective tissue that lets a general coding agent orchestrate flows natively and fill the breaks; this paper asks which factor bottlenecks such an agent, and at which layer.

We present **ZhuLong**, an execution-grounded LLM coding agent for PyAether, SKILL, and Tcl: four unified MCP tools [13]—semantic and keyword API search, documentation lookup, and sandbox execution—close a generate–execute–refine loop over four commercial runtimes via a single `lang–host` pair, with offline **API self-exploration** as a bounded complement. We also give a **theoretical grounding** (Section 4): EDA correctness is situated (L4), a layer that offline knowledge—and hence model scale—cannot reach; we argue that the grounding gap on L4 failures is bounded by observation fidelity.

Contributions.

- **RQ1 (binding constraint).** We formalize the grounding gap and measure both axes it depends on: with the LLM fixed, the system-level observation axis (pure LLM, no observation → full system) spans 57.2 pp; with the harness fixed, the backbone axis spans 23.7 pp. The two axes answer different questions, making observation the binding constraint in the current model-capability regime, while model scale still sets the L1–L3 ceiling (Section 4, 6).
- **RQ2 (necessary components).** Ablation isolates retrieval, execution, and belief update as jointly necessary, with toolchain-specific magnitudes that make them complementary rather than interchangeable; the cost columns add that full-fidelity read-back is also the cheaper feedback to operate (Section 6).
- **RQ3 (toolchain generality). EDA-Scripting-Bench** (449 tasks, 4 runtimes, one schema) re-measures the same structure slice by slice; we specify the harness, schema, and assertion protocol in full (Section 5).

2 Background and Motivation

We target three widely-used EDA scripting interfaces across four commercial runtimes: PyAether for Empyrean Aether [7], SKILL for Cadence Virtuoso [16], and Tcl for Synopsys Custom WaveView and Cadence Innovus. Despite their prevalence, scripting for these environments poses challenges beyond what generic LLM code generators can address.

EDA scripting is fundamentally different. Prior LLM code generation succeeds where correctness is verified by unit tests [1, 3]. EDA scripting presents three distinct challenges: (i) APIs are long-tailed, tool-specific, and poorly documented; (ii) correctness depends on side effects in a live, stateful database, not return values; and (iii) the community lacks a standardized benchmark. The first two are technical; the third is infrastructural. Challenge (ii) is the one that resists a purely offline treatment: a script can be syntactically well-formed and invoke documented APIs yet still fail because the live design state it acts on is not what the offline knowledge assumed.

Prior custom/analog scripting work stops short of execution. LayoutCopilot [11] targets analog *layout* (not scripting); ChaTCL [15] and RAG-EDA [14] generate Tcl scripts from static knowledge bases without executing them, so runtime errors go undetected. Execution feedback is not new—in the digital flow, ChatEDA [8], MCP4EDA [18], and AutoEDA [13] close the loop against synthesis and P&R state on open toolchains (Yosys, OpenLane, OpenRoad)—but none grounds against the live layout, schematic, or waveform database of custom/analog tools, the surface this paper targets. A recent survey [12] organizes EDA

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agents by handoff validity; ZhuLong targets the custom/analog scripting handoff, previously ungrounded, and our benchmark is the first for this setting (concurrent benchmarks focus on digital Tcl [19] or physical-design agents [10]).

Our approach. ZhuLong closes the loop from generation to execution to refinement, supplies a **theoretical grounding** (Section 4) explaining *why* execution is necessary—it is the only channel that observes the current design state (L4), which offline knowledge cannot supply—and provides the missing infrastructure: **EDA-Scripting-Bench** (Section 5).

3 System Design

3.1 Overall Architecture

ZhuLong extends Cline-CLI [4] with EDA-specific retrieval, documentation, and execution. The runtime is an **ICL-type autonomous agent**: it decides its own next action, keeps no explicit posterior, and maintains its belief state implicitly via in-context learning (Section 4). The policy stays commodity and unmodified, so gains are attributable to the interface, and the runtime's permission/hook mechanisms supply the enforcement points the fixed protocol relies on (Section 6.1). As in Figure 1, three components—the LLM agent runtime, the API knowledge base, and the EDA execution environment—are linked through four MCP tools, with offline **self-exploration** pre-augmenting the knowledge base.

3.2 Four MCP Tools

The four tools jointly realize the observation channel Ω and the effector of Section 4, each taking a `lang–host` pair that selects the runtime so one agent logic covers all four: `search_apis` retrieves APIs by semantic similarity (BGE-M3 embeddings [2] with FAISS [6], top-K, merged with BM25); `search_apis_by_keyword` retrieves by keyword (CamelCase plus BM25); `get_api_details` fetches documentation (parameters, returns, examples), augmented by self-exploration; and `run_code` executes code in an isolated headless sandbox and reads back a structured result (status, error log, stdout, return) for refinement, dispatching to resident or one-shot workers per runtime.

3.3 Agent Workflow

The agent iteratively refines its output through execution feedback [4]: reasoning over the instruction, retrieving and inspecting candidate APIs, generating and executing code, then re-planning based on the observed result. The agent is not constrained to a fixed sequence—it may skip retrieval when it already knows the API or conduct multiple rounds when initial results lack confidence—and terminates when it judges further attempts unlikely to succeed.

3.4 Offline API Self-Exploration (Complementary)

EDA API documentation is often incomplete, omitting parameter constraints, return structures, and error conditions. Per Section 4, retrieval enriches only the L1–L3 facts Ω returns, so self-exploration complements—and is bounded by—the L4 observation; it is an efficiency mechanism, not the accuracy driver. Offline, the exploration agent reads documentation, identifies gaps, generates test scripts, executes them via `run_code`, and iteratively refines a behavioral model of constraints, return structures, error conditions, and side effects. Exploration stops when a per-API confidence score exceeds threshold τ, a round adds no new constraint, or a per-API budget M is exhausted; scripts are expert-gated to filter destructive operations, and the result is served at runtime via `get_api_details` with no overhead.

4 Theoretical Grounding: The Grounding Gap

This section develops the *grounding gap* framework, stated before the measurements of Section 6 so that its claims are refutable predictions rather than fitted explanations: EDA correctness is a *situated* property that cannot be guaranteed from offline knowledge, execution is the only channel that observes the current design state, and on the L4 failures that offline knowledge cannot reach, the resulting gap is bounded by observation fidelity—while model scale continues to set the L1–L3 ceiling.

4.1 A Four-Level Hierarchy of Information

To separate what can be learned offline from what must be observed online, Table 1 distinguishes four levels: L1 *syntactic correctness*, L2 *physical plausibility*, and L3 *historical factuality* (documented, past-oriented, frozen at the training cutoff) are learnable offline, whereas L4 *situated truthfulness*—what is true *right now* in a specific environment—is not, defined by indexicality (truth depends on context and changes over time), non-guaranteeability from offline data, and real-time commitment. L4 must be *read* from the environment, not *recalled* from memory.

For EDA scripting, the situated state is the live design database—which cell/view is selected and what the layout or schematic contains now—plus the user's current intent (a user clarification is also an L4 observation). Our correctness criterion—code runs without error and all assertions over the mutated design pass—is L4: a well-formed (L1), plausible (L2) script invoking documented functions (L3) can still violate an undocumented binding contract, depend on internal selection state, or target the wrong cell. Such failures cannot be resolved offline; they must be *observed* by acting on the live design and reading back the result.

4.2 A Bayesian Formalization of the Grounding Gap

We cast grounding as decision-making under partial observability [9, 17]; the formulation is diagnostic rather than algorithmic—it makes precise what is missing (the observation likelihood Ω and the recursive update Φ) without prescribing tractable inference.

Let Θ denote the space of L4 states (the live EDA design database), o_t a runtime observation, a_t a generated code action, and $\mathcal{K}$ the offline knowledge (L1–L3) in the LLM's frozen parameters. The agent maintains a belief $b_t(\theta) \triangleq p(\theta_t \mid o_{1:t}, a_{1:t-1}, \mathcal{K})$ over the current L4 state, updated recursively by Bayes' rule:

$$b_t(\theta) \propto p(o_t \mid \theta_t) \int_\Theta p(\theta_t \mid \theta_{t-1}, a_{t-1})\, b_{t-1}(\theta_{t-1})\, d\theta_{t-1}, \quad (1)$$

where $p(o_t \mid \theta_t)$ is the *observation likelihood* Ω and $p(\theta_t \mid \theta_{t-1}, a_{t-1})$ is the *transition model* $\mathcal{T}$ supplied by the LLM's prior;

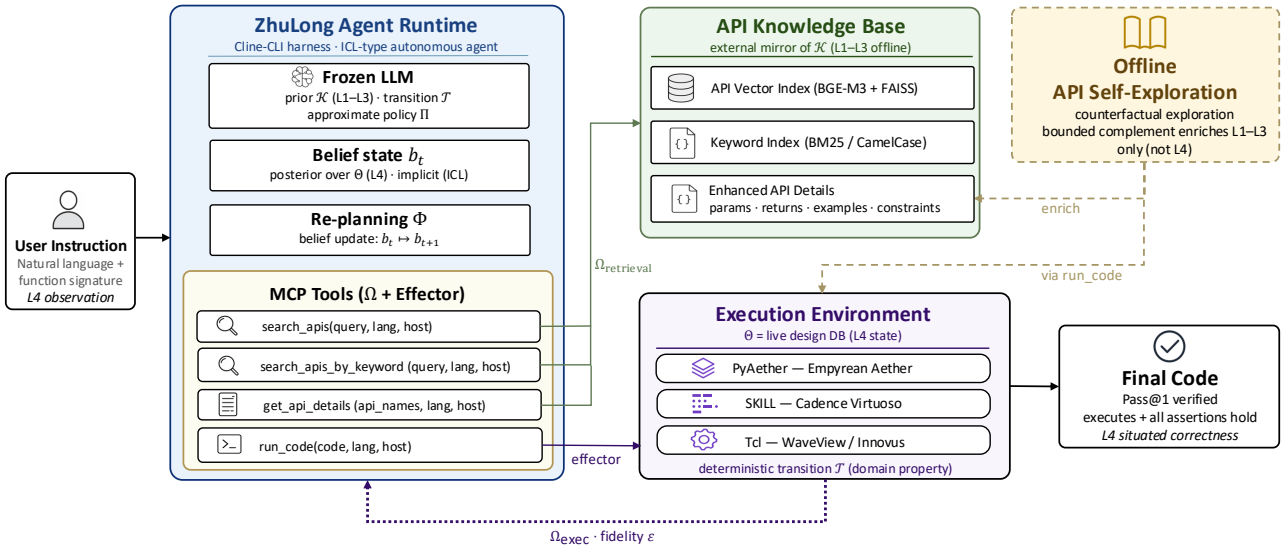


Figure 1: ZhuLong system architecture. The LLM agent runtime, the API knowledge base, and the EDA execution sandbox are linked through *four* unified MCP tools that dispatch on a single `lang`–`host` pair to *three* platforms (PyAether, SKILL, Tcl). Offline API self-exploration pre-augments the knowledge base (dashed); execution feedback is looped back for iterative refinement.

Table 1: The four levels of information

Level	Name	Core Question	Learnable Offline?
L1	Syntactic Correctness	Is this string well-formed?	Yes
L2	Physical Plausibility	Could this event happen?	Yes
L3	Historical Factuality	Did this event happen?	Yes
L4	Situated Truthfulness	Is this true *right now*?	**No**

actions are chosen by an approximate policy Π that minimizes expected cost under b_t, with $\mathcal{C}$ encoding the task assertions. Eq. (1) is the standard POMDP filter [9]; what is new here is not the filter but the *implementation* of Ω as retrieval plus execution read-back, which is this work's contribution to the POMDP stack.

The *grounding gap* is the divergence between the posterior achievable with runtime observations and the posterior from the offline prior alone:

$$D_{\text{grounding}}(t) \triangleq D_{\text{KL}}(p(\cdot \mid o_{1:t}, \mathcal{K}, \text{env}) \parallel p_{\mathcal{K}}(\cdot)), \qquad (2)$$

where the right-hand side is the *pure prior* unconditioned on any observation. We now bound this gap.

Scenario: the transition is deterministic. In EDA, `run_code` executes a script whose effect on the design database is deterministic: the same action in the same state yields the same result, so $\mathcal{T}$ is exact—a property of the domain, not a simplifying assumption. The only mismatch is in the observation likelihood Ω: retrieval may return incomplete documentation, and execution read-back may truncate—precisely what ϵ captures. Model mismatch under estimated parameters is a recognized obstacle [5]; our setting is milder, with only Ω, not $\mathcal{T}$, possibly mismatched.

Claim (observation-fidelity bound). Let $\epsilon \triangleq \sup_\theta \|p(\cdot \mid \theta) - \hat{p}(\cdot \mid \theta)\|_{\text{TV}}$, the total-variation distance between the true and the agent-used observation likelihoods. We posit that $D_{\text{grounding}} \leq g(\epsilon)$, with $g(\epsilon) \to 0$ as $\epsilon \to 0$ and g *independent of the complexity of the prior $\mathcal{K}$ on task distributions where L4 failures dominate.* That is, on the failures offline knowledge cannot reach, the grounding gap is bounded by observation fidelity; on L1–L3 failures, the prior $\mathcal{K}$ remains the relevant lever, so a stronger model raises that ceiling without substituting for observation. We state this as a testable *prediction*; the next paragraph tests it from both sides.

Experimental correspondence (developed in Section 6). Here ϵ is exactly the fidelity of this paper's Ω implementation—retrieval quality plus execution read-back. The read-back ladder (N′ → B → F, Table 3) varies ϵ with retrieval and the effector fixed; the backbone axis (Table 4) varies $\mathcal{K}$ with ϵ fixed. The two axes answer different questions—how much observation buys at a fixed model, versus how much a stronger prior buys under

full observation—so we report both and ask which is binding in the current model-capability regime, rather than treating them as competing explanations. The read-back ladder isolates ϵ and is monotone where tasks are not near-single-shot (Tcl excepted); the system-level ϵ axis spans 57.2 pp while the backbone axis spans 23.7 pp, which places observation as the binding constraint here without implying it dominates at all scales or on all task distributions.

4.3 Design Principles

The formalization yields three architectural principles. **Decouple knowledge from belief:** adapt in the dynamic posterior b_t the Harness maintains, not by updating the frozen prior $\mathcal{K}$. **Optimize observation, and recognize where the prior saturates:** on L4 failures the gap is bounded by Ω's fidelity ($g(\epsilon)$ on the distributions where L4 dominates, independent of model scale), so observation investment pays directly there—full observation beats binary feedback on both accuracy and operating cost (Section 6.2)—while a richer prior raises the L1–L3 ceiling with diminishing returns on L4. This is a bottleneck, not a substitution: neither axis compensates for the other. **Design for continuous belief updating:** Φ runs online ($b_t \mapsto b_{t+1}$); in the ICL-type agent (Section 3) what is bounded is not a round count but how much evidence belief can accumulate—a graded execution budget or a lagging read-back.

5 Benchmark Construction

5.1 Dataset: EDA-Scripting-Bench

EDA-Scripting-Bench is the instrument for both of this paper's comparisons: within one toolchain, the effect of each grounding component, and across toolchains, the effect of language structure. It covers PyAether, SKILL, and Tcl on four commercial runtimes, **449 tasks** in four slices (Table 2), all specified by one schema—a target language and runtime, a natural-language prompt, a function signature, and language-appropriate assertions—so a single harness (Section 3.3) scores all four platforms uniformly. Scoring is execution-based (a task passes only when the generated code acts correctly in the live runtime), and the cross-language slices are difficulty-controlled by construction (Section 5.3). Each task is a real break in a design flow where a closed tool stops and glue must be hand-written.

Table 2: Composition of EDA-Scripting-Bench

Slices and runtimes		
Slice	**Runtime**	**Tasks**
EDA-Eval-PyAether	Empyrean Aether	158
EDA-Eval-Skill	Cadence Virtuoso	157
EDA-Eval-WaveViewTcl	Synopsys Custom WaveView	77
EDA-Eval-InnovusTcl	Cadence Innovus	57
Total		**449**

5.2 The PyAether slice (EDA-Eval-PyAether)

The reference slice, **EDA-Eval-PyAether** (158 tasks), anchors every task to a concrete PyAether API; EDA experts curated each assertion to be self-contained and executable in a clean sandbox without vendor-internal context. Tasks span layout, schematic, and design-management workflows, compose several distinct APIs, and sometimes depend on a technology library, so passing reflects command of the tool's behavior, not surface retrieval.

5.3 SKILL and Tcl slices

The remaining slices are built so cross-language scores are comparable. **EDA-Eval-Skill** (157 SKILL tasks on Cadence Virtuoso) is *function-aligned* with PyAether: each task re-implements the same entry point and functional specification as its PyAether counterpart, holding difficulty fixed. The **Tcl** slices are graded by difficulty across two further toolchains: waveform measurement on WaveView (**EDA-Eval-WaveViewTcl**, 77 tasks) and floorplanning, placement, and routing on Innovus (**EDA-Eval-InnovusTcl**, 57 tasks), so the cross-language comparison (Section 6.2.4) attributes score differences to language structure rather than task hardness.

5.4 Evaluation Protocol

A task passes if the generated code executes in its target runtime sandbox without errors and all assertions hold; the primary metric is Pass@1 [3], reported as mean ± s.d. over five independent runs (reshuffled task orderings, fresh agent contexts), per slice and pooled over all 449 tasks.

6 Experiments

6.1 Experimental Setup

Research questions. The framework of Section 4 makes three testable claims, one per RQ: **RQ1 (binding constraint)**—which axis, observation fidelity Ω or the prior $\mathcal{K}$, is binding, and over what range? (S1 varies Ω, S3 varies $\mathcal{K}$; the two axes answer different questions and are reported as such.) **RQ2 (necessary components)**—do observation, execution, and belief update each matter, and can one substitute for another? (S2 varies the effector and update budget.) **RQ3 (toolchain generality)**—does the structure reproduce on another toolchain? (S4 re-runs S1–S2 on SKILL and Tcl.)

Benchmark and protocol. We evaluate on **EDA-Scripting-Bench** (449 tasks, Section 5) using **Pass@1**: one trace per task (the agent may iterate within it via Φ but is never restarted), a 2500-second timeout, and a `PreToolUse` hook that enforces a tool whitelist, blocks shell/code-search tools, and confines file access to a fresh directory—identical in every configuration. Each configuration runs five times (reshuffled order, fresh contexts), reported as mean ± s.d. Task cases derive from non-redistributable vendor documentation (unlikely in public pretraining), and no third-party agent can run these runtimes without the MCP bridge, so baselines are tool-registration ablations of the same harness—enabling single-variable attribution.

Configurations. We isolate components by tool registration (a disabled tool is unregistered and invisible; the agent policy is held fixed), giving five configurations (Table 3): *Pure LLM* (no retrieval, sandbox, or self-exploration), *RAG* (retrieval + self-exploration, no sandbox; = w/o-Sandbox), *w/o Self-Expl.* (retrieval + sandbox), *w/o Retrieval* (sandbox + self-exploration), and *ZhuLong (Full)* (all three

Table 3: Main results on EDA-Scripting-Bench: Pass@1 (%) over five runs (mean ± s.d.), pooled over all 449 tasks and per slice. The five right-most columns report per-task operating cost over all 449 tasks (five runs, mean ± s.d.). The lower block varies the *read-back* Ω alone (retrieval and effector fixed); (F) coincides with the Full row.

	Pass@1 (%)				Per-task operating cost (449 tasks)				
Configuration	Overall (449)	PyAether (158)	SKILL (157)	Tcl (134)	Time (s)	run_code	Tool calls	In tok (k)	Out tok (k)
Pure LLM	19.7±1.4	14.3±2.1	33.2±2.3	10.1±1.1	183±9	—	3.6±0.1	72±2	6.3±0.7
RAG (retr.+self-expl., no sandbox)	68.6±1.6	71.1±4.6	52.2±1.9	84.6±1.9	226±9	—	16.9±0.7	272±8	8.6±0.3
ZhuLong w/o Retrieval	66.2±4.2	73.4±7.3	45.2±9.4	82.2±2.3	981±151	22.6±2.8	26.3±3.1	658±133	11.7±2.8
ZhuLong w/o Self-Expl.	74.5±1.8	77.0±4.1	**61.5±0.7**	86.7±3.0	645±132	12.6±1.4	29.5±2.9	722±116	11.5±2.3
ZhuLong (Full)	**76.9±2.3**	**82.5±2.0**	60.3±6.7	**89.9±2.0**	464±57	10.6±0.7	26.3±1.4	710±21	12.0±0.7
Read-back axis (retrieval + effector fixed, only what run_code *returns varies)*									
(N′) No read-back (prior only)	69.8±2.2	73.4±2.1	50.7±4.9	87.9±1.7	253±5	1.1±0.0	16.6±0.9	278±12	4.0±0.1
(B) Binary pass/fail	71.0±0.8	74.6±1.6	57.2±4.8	83.0† ±2.3	634±40	22.8±2.9	43.9±3.9	887±139	7.7±0.8
(F) Full observation (= Full row)	76.9±2.3	82.5±2.0	60.3±6.7	89.9±2.0	464±57	10.6±0.7	26.3±1.4	710±21	12.0±0.7

Bold marks the per-column best. — = no sandbox execution. † On Tcl, read-back inverts at the middle rung (short near-single-shot scripts solved largely from documentation; see Section 6.2.1). The Tcl column pools the two Tcl hosts, WaveView (77 tasks) and Innovus (57), by summing pass counts per run over 134 tasks. Time = net-active solve time (final-session active-seconds, excluding quota wait); run_code = execution calls per task; Tool calls = total tool invocations per task (search + details + execution + builtin); Input/Output tokens = input/output tokens per task (thousands), each averaged over all 449 tasks (five runs). Read-back-axis rows (N′, B) are costed on the same per-task, 449-task caliber.

on). S1 probes the observation axes at their extremes; S2 bounds Φ. All main experiments use DeepSeek-V4-Pro as the backbone; S3 evaluates five additional backbones under the identical harness.

6.2 Results and Ablations

Table 3 gives the five configurations pooled and per slice. Relative to the full system, removing execution costs 8.3 pp pooled, removing retrieval 10.7 pp, and removing self-exploration 2.4 pp; against the pure-LLM floor the full system gains 57.2 pp. Wilson 95% intervals on the pooled Pass@1 (n=2245): Full [75.11, 78.60], w/o-Self-Expl. [72.66, 76.26], w/o-Retrieval [64.22, 68.13], RAG [66.65, 70.49], N′ [67.87, 71.66], B [69.09, 72.84], Pure-LLM [18.11, 21.40]; the controlled $N' \to F$ read-back gap of 7.1 pp has 95% interval [4.52, 9.68] pp (SE= 1.32 pp). The same approximation gives the two RQ2 component gaps, which exclude 0—retrieval +10.7 pp [8.1, 13.3] and execution +8.3 pp [5.7, 10.9]—while self-exploration +2.4 pp [−0.1, 4.9], the binary-over-none read-back rung +1.2 pp [−1.5, 3.9], and the SKILL self-exploration difference −1.2 pp [−3.6, 6.0] straddle 0, so effects of this size are below this benchmark's power to certify, consistent with their complementary framing.[1]

6.2.1 S1: Observation fidelity (RQ1). Principle 2 predicts that with the model fixed, accuracy is bounded by the fidelity of Ω *on the failures that require observing the live design.* **Documentation axis.** Holding the agent loop, effector, and LLM fixed, removing retrieval costs 10.7 pp pooled, but highly toolchain-specific: on PyAether the read-back channel partially substitutes for retrieval (sandbox introspection via `dir()`/`inspect`/`__doc__` rises from 49.0% to 86.7% of traces once retrieval is removed, reconstructing the API knowledge retrieval would return). Tcl substitutes likewise through its own enumeration channels—WaveView's `info commands` returns the full command surface in one call, and Innovus's `get_db` probing enumerates attributes by trial and error—recovering the API surface but not its deep semantics, so removing retrieval there costs only 7.7 pp. SKILL's sandbox has no such enumeration channel (`apropos`/`describe` fail), leaving only `boundp`/`isCallable` existence checks on guessed names, so the API surface cannot be rediscovered and retrieval there is irreplaceable (15.1 pp). Retrieval is thus one *implementation* of Ω, not the channel itself.

Read-back axis. The second axis is how much of the live state execution returns. We compare three closed-loop variants differing only here (Table 3, lower block): (N′) *no read-back*—`run_code` executes but returns nothing; (B) *binary feedback*—pass/fail only; (F) *full observation*—stdout, traces, per-assertion outcomes, and queried state (the default). The predicted monotone ordering N′ < B < F holds pooled and on the two largest slices, though on PyAether the middle rung is narrow (+1.2 pp, within one combined s.d.): a bare pass/fail verdict adds little actionable information, and the F rung carries almost the whole ladder (+7.9 pp of the 9.1 pp N′-to-F gap). On Tcl it inverts at the middle rung—short, near-single-shot scripts solved from documentation leave a binary signal nothing to act on. Cost sharpens the reading (Table 3): binary feedback is *more* expensive than full observation in time, `run_code` calls, input tokens, and total tool calls (634 vs. 464 s, 22.8 vs. 10.6, 887k vs. 710k, and 43.9 vs. 26.3 per task), emitting fewer output tokens (7.7k vs. 12.0k)—without actionable error information the agent probes blindly—while no read-back collapses cost (253 s, 1.1 calls) together with accuracy. What the channel returns, not how often the agent acts, drives both—the N′-to-F gap operationalizes the grounding gap as an accuracy difference.

Self-exploration rung. The top rung adds constraints discovered offline (Section 3.4); per the framework this enriches only L1–L3 facts, so it should move the result less than the channel's fidelity. Removing it (retrieval and effector kept) drops Pass@1 from 82.5% to 77.0% on PyAether (76.9% to 74.5% pooled), a bounded prior-side complement rather than a substitute. Self-exploration is cost-negative at runtime (cutting net-active solve time by 28.1%, see Table 3), whereas removing retrieval makes the agent fall back on sandbox introspection (2.1× the full system's time).

6.2.2 S2: The closed loop (RQ2). Principle 3 requires that actions reach the design through a causal interface and that Φ run continuously. **Effector.** Removing the sandbox drops pooled Pass@1 from 76.9% to 68.6% (−8.3 pp): without an effector the agent becomes a prior-only predictor. **Belief update.** We bound how often

[1] All difference intervals treat the two arms as independent and pool task-runs as i.i.d., both of which overstate variance; a paired, task-clustered analysis would only tighten them and strengthen the headline claims.

Table 4: Impact of LLM backbone (ZhuLong, full harness).

Backbone	PyAether (158)	SKILL (157)	Tcl (134)	Overall (449)	Δ
DeepSeek-V4-Pro	82.5±2.0	**60.3±6.7**	**89.9±2.0**	**76.9±2.3**	–
GLM-5.3-Flash	**86.8±1.6**	57.2±6.9	80.3±3.8	74.5±2.4	−2.4
GLM-5.2	83.3±3.1	54.4±4.5	82.7±1.6	73.0±2.1	−3.9
Kimi-K2.6	77.0±1.6	55.5±2.7	75.8±1.9	69.1±0.5	−7.8
DeepSeek-V4-Flash	60.1±5.5	47.0±1.8	70.4±3.1	58.6±2.3	−18.3
Doubao-Seed-2.0-Pro	63.8±1.4	42.5±3.4	53.3±3.4	53.2±1.9	−23.7

Pass@1 (%), five runs (mean ± s.d.); Tcl pools WaveView (77) and Innovus (57), by summing pass counts per run over 134 tasks; Δ vs. the primary backbone, pooled. Bold = per-column best. The strongest-to-weakest span is 76.9 − 53.2 = 23.7 pp.

the agent *may* execute (a per-trace `run_code` budget, denied at the tool boundary) rather than forcing early submission, and vary the evidence along two axes: the budget k, and a *lagged* channel that returns the *previous* execution's state, leaving the agent one action behind the design. On PyAether, accuracy rises monotonically with the budget (63.8% at k=1, 69.1% at k=3, 77.7% at k=10, unsaturated); the lagged channel lands between k=3 and k=10 (73.0%)—stale evidence is worth markedly less than fresh—and tighter budgets leave traces unfinished (Converged drops from 99.1 to 76.3 at k=1).

6.2.3 S3: The prior axis (the prior-side contrast of RQ1). Principle 1 separates knowledge from belief: the frozen prior sets the L1–L3 ceiling a harness can reach without substituting for observation. Table 4 changes only $\mathcal{K}$ (the LLM) with the harness fixed; the backbone remains decisive—even with retrieval, sandbox, and self-exploration in place, the strongest-to-weakest spread is 23.7 pp pooled. This spread is what the $\mathcal{K}$ axis can move (Principle 2): it raises the L1–L3 ceiling but cannot reach L4, so it does not substitute for observation. The system-level observation axis spans 57.2 pp against a 23.7 pp backbone range—two axes answering different questions, making observation the binding constraint in the current model-capability regime, without implying that it dominates at all scales or on all task distributions. GLM-5.3-Flash beats the primary on PyAether yet loses 2.4 pp pooled—slice-level snapshots can invert backbone rankings, itself an argument for the pooled protocol.

6.2.4 S4: The toolchain axis (RQ3). The four slices share one schema, assertion criterion, and harness, so S1–S2 re-runs slice by slice with nothing changed but the toolchain. Table 3 carries the comparison: the pure-LLM floor and the closed-loop advantage reproduce on every slice (except SKILL, where Full trails w/o-Self-Exploration by 1.2 pp), and the read-back ordering N′ < B < F holds on the two largest slices and pooled (narrowly at the middle rung on PyAether; Tcl inverts, see Section 6.2.1). Component magnitudes are toolchain-specific—execution buys +8.1 pp on SKILL but only +5.3 on Tcl, and retrieval buys +15.1 pp on SKILL, +9.1 pp on PyAether, and +7.7 pp on Tcl, tracking each host's introspective surface—but the structure (observation and execution jointly closing the loop) reproduces on every slice.

6.3 Error Analysis

ZhuLong passes 82.5% on PyAether (652/790 task-rounds), 60.3% on SKILL (473/785), 88.6% on WaveView (341/385), and 91.6% on Innovus (261/285; Table 3). We split each slice's misses into a *generation-stage* failure (no executable candidate returned—search budget saturated or final code lost) and an *evaluation-stage* failure (a candidate returned but earning no pass). The generation/evaluation split is 29/109 on PyAether, 31/281 on SKILL, 1/43 on WaveView, and 1/23 on Innovus.

Across all four slices, the evaluation-stage failures that admit a clean attribution fall into four recurring patterns—all on solvable tasks ZhuLong misreads through an underspecified or invisible contract: **(i) Wrong internal API or contract**, a plausible but incorrect long-tail API, name, or field literal that survives until execution rejects it; **(ii) Unauthorized defensive guards**, a self-added validation or failure path that rejects input the specification never excluded; **(iii) Underspecified boundary and representation semantics**, an ambiguity the prompt leaves open that is resolved on the wrong side, flipping the verdict across runs; and **(iv) Host language and context contracts**, a type-template, headless-context, or canonical-form convention violated so the candidate dies before judging.

Concretely, these patterns surface as constraints documented nowhere and only weakly exposed by execution: PyAether-068 builds the cartoon-box highlight through the plausible-but-wrong `aeAddHighLightRect` where the oracle requires `GuCartBox(emyBoxF(...))`; WaveView-037 returns `N/A` on a single-rising-edge waveform and SKILL-037 rejects an empty hierarchy path—guards the specification never granted; Innovus-057 over-filters through an extra `.use=="signal"` predicate to report 1869 weight-2 nets against the oracle's 1872; and SKILL-008 hardcodes the schematic view type when opening a layout view, missed by every evaluated run although a direct probe proves the task passable.

7 Discussion

Section 4 elevates the error analysis of Section 6.3 from observation to structural principle: in domains with long-tailed, under-documented APIs, execution channels are the binding constraint on L4 failures—those offline knowledge, hence a larger prior, cannot reach—while a stronger prior raises the L1–L3 ceiling without substituting for it. On the L4 failures that dominate here, full observation is both more accurate and cheaper than binary feedback (Section 6.2.1).

Three limitations. Base-LLM performance varies widely (53.2%–76.9% pooled)—consistent with the framework (the $\mathcal{K}$ axis moves L1–L3)—and the benchmark size limits power to certify small effects, so the self-exploration gain (+2.4 pp) and the binary-over-none read-back rung (+1.2 pp) are complementary rather than headline; self-exploration also costs a one-time pass per API, and the interactive bridge is single-threaded. Per-slice dispersions can be large (e.g., SKILL w/o Retrieval ±9.4), so slice-level comparisons need care, and the SKILL self-exploration regression (−1.2 pp) fits SKILL's smaller API surface—offline facts are often redundant with retrieval and crowd out task-relevant evidence. Pass@1 certifies only the effects its assertions check, not full functional equivalence. The system is deployed in production EDA flows, but deployment metrics and full details cannot be disclosed here.

8 Conclusion and Future Work

ZhuLong reframes EDA scripting as the connective tissue that makes design flows runnable end-to-end: a general coding agent orchestrates flows natively and, where closed EDA tools leave gaps, synthesizes the missing glue. The execution-grounded loop and the grounding-gap framework explain why: on L4 failures model scale cannot reach, the gap is bounded by observation fidelity, while model scale still sets the L1–L3 ceiling. The full system reaches **76.9%** pooled Pass@1. Future work: (i) domain-specializing the harness, (ii) long-term memory and recursive self-improvement (RSI), (iii) multimodal grounding, and (iv) benchmarking design-flow orchestration.

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